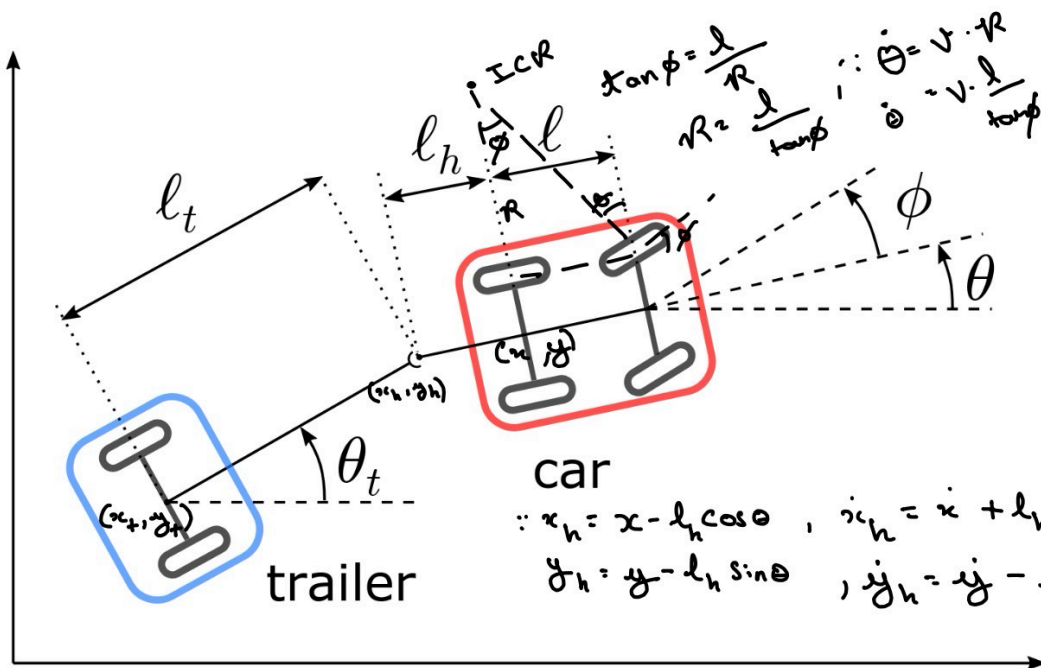




car

$$\therefore x_h = x - l_h \cos \theta, \quad \dot{x}_h = \dot{x} + l_h \dot{\theta} \sin \theta$$

$$y_h = y - l_h \sin \theta, \quad \dot{y}_h = \dot{y} - l_h \dot{\theta} \cos \theta$$

— ①

$$x_t = x_h - l_t \cos \theta_t, \quad \dot{x}_t = \dot{x}_h + l_t \dot{\theta}_t \sin \theta_t = \dot{x} + l_h \dot{\theta} \sin \theta + l_t \dot{\theta}_t \sin \theta_t$$

$$y_t = y_h - l_t \sin \theta_t, \quad \dot{y}_t = \dot{y}_h - l_t \dot{\theta}_t \cos \theta_t = \dot{y} - l_h \dot{\theta} \cos \theta - l_t \dot{\theta}_t \cos \theta_t$$

Assignment 3 - Wheeled Mobile Robots

Exercise 1:

State variable $[q]$:

x, y : global coordinates of the Car's Rear Axle.

θ : Heading of the Car.

θ_t : Heading of the Trailer.

Control inputs $[u]$:

v : velocity of the Car's rear wheels

ϕ : The steering angle of the front wheels.

ω : Steering speed $[\dot{\phi} = \omega]$

Lengths:

l : length of the car

l_h : Distance from the rear axle to the connected point

l_t : length of the trailer.

Now, for the kinematics of the car, we define the motion of the car using the Bicycle Model.

Rear Axle velocity $[\dot{x}, \dot{y}]$ (no sliding):

$$\dot{x} = v \cos \theta$$

$$\dot{y} = v \sin \theta$$

①

Car Angular velocity $[\dot{\theta}]$, $\dot{\theta} = \frac{v}{l} \cdot \tan \phi$

②

Now, for the kinematics of the Trailer:

● P_h : $x_h = x - l_h \cos \theta$
 $y_h = y - l_h \sin \theta$ — (3)

Now for the velocity of the $P_h [\dot{P}_h]$

∴ $\dot{x}_h = \dot{x} + l_h \dot{\theta} \sin \theta$, from (3)

$\dot{y}_h = \dot{y} - l_h \dot{\theta} \cos \theta$
 — (4)

Now for the position of the trailer wheels $[P_t]$

∴ $x_t = x_h - l_t \cos \theta_t$

$y_t = y_h - l_t \sin \theta_t$ — (5)

Now for the velocity of the trailer wheels $[P_t]$

from (5)

∴ $\dot{x}_t = \dot{x}_h + l_t \dot{\theta}_t \sin \theta_t$

$\dot{y}_t = \dot{y}_h - l_t \dot{\theta}_t \cos \theta_t$ — (6)

∴ from (4) and (6)

$\dot{x}_t = [\dot{x} + l_h \dot{\theta} \sin \theta] + l_t \dot{\theta}_t \sin \theta_t$

$\dot{y}_t = [\dot{y} - l_h \dot{\theta} \cos \theta] - l_t \dot{\theta}_t \cos \theta_t$, from (1)

$\dot{x}_t = v \cos \theta + l_h \dot{\theta} \sin \theta + l_t \dot{\theta}_t \sin \theta_t$

$\dot{y}_t = v \sin \theta - l_h \dot{\theta} \cos \theta - l_t \dot{\theta}_t \cos \theta_t$ — (7)

Now applying the constraint Equation [Trailer wheels cannot slide sideways]:

$$\dot{x}_t \sin \theta_t - \dot{y}_t \cos \theta_t = 0 \quad \text{--- (8)}$$

from (7) (8)

$$[v \cos \theta + l_h \dot{\theta} \sin \theta + l_t \dot{\theta}_t \sin \theta_t] \sin \theta_t - [v \sin \theta - l_h \dot{\theta} \cos \theta - l_t \dot{\theta}_t \cos \theta_t] \times \cos \theta_t = 0$$

Rearranging:

$$v [\cos \theta \sin \theta_t - \sin \theta \cos \theta_t] + l_h \dot{\theta} [\sin \theta \sin \theta_t + \cos \theta \cos \theta_t] + l_t \dot{\theta}_t [\sin^2 \theta_t + \cos^2 \theta_t] \quad \text{--- (9)}$$

Using trigonometric identities:

$$\therefore \sin A \cos B - \cos A \sin B = \sin [A - B]$$

$$\cos A \cos B + \sin A \sin B = \cos [A - B]$$

$$\sin^2 A + \cos^2 A = 1$$

$$\text{--- (10)}$$

from (9) (10)

$$v \sin [\theta - \theta_t] + l_h \dot{\theta} \cos [\theta - \theta_t] + l_t \dot{\theta}_t = 0$$

$$\therefore \dot{\theta}_t = \frac{v}{l_t} \sin [\theta - \theta_t] - \frac{l_h}{l_t} \dot{\theta} \cos [\theta - \theta_t]$$

from (2)

$$\dot{\theta}_t = \frac{v}{l_t} \sin [\theta - \theta_t] - \frac{l_h}{l_t} \left[\frac{v}{l} \tan \phi \right] \cos [\theta - \theta_t]$$

$$\therefore \dot{\theta}_t = v \left[\frac{1}{l_t} \cdot \sin[\theta - \theta_t] - \frac{l_h}{l_t} \cdot \cos[\theta - \theta_t] \cdot \tan\phi \right] \quad \text{--- (11)}$$

Now for the final Kinematic Model

$$\therefore \text{State vector } z: q, [x, y, \theta, \theta_t, \phi]^T$$

The Kinematic Model :

from (12) and (11)

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\theta}_t \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \\ \frac{1}{l} \tan\phi \\ \frac{1}{l_t} \cdot \sin[\theta - \theta_t] - \frac{l_h}{l_t} \cos[\theta - \theta_t] \tan\phi \\ 0 \end{bmatrix} v + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \omega \quad \text{--- (12)}$$